$f: M \to M$ → surjection (injection) → f: iso

↑ Noetherian (Artin~)

Artinian ring

dcc

$\mathbb{Z}$ $(2) \supset (2^2) \supset (2^3) \supset \cdots$

$M/\ker f \cong M$

$\ker T^n \cap \mathrm{Im}\, T^n$

$\ker T^n$

$0 \to L \xrightarrow{f} M \xrightarrow{g} N \to 0$

$0 \to L \xrightarrow{f} M$ ⇔ f: inj

$M \xrightarrow{g} N \to 0$ ⇔ g: sur

M: 노ether ⇔ L, N: 노ether

(Artin) (Artin)

$M/f(L) \cong N$

$M \cong N$

Cor M_i: 노ether $(1 \le i \le n)$

$\Rightarrow \bigoplus_{i=1}^{n} M_i$: 노ether

$0 \to M_1 \to \bigoplus_{i=1}^{n} M_i \to \bigoplus_{i=1}^{n-1} M_i \to 0$.

$T: V \to V$ is diagonalizable $\Leftrightarrow$ m_T splits and separable

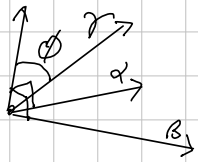
" triangluriazable $\Leftrightarrow$ m_T is separable

relation {{

E/F is separable $\Leftrightarrow$ $\#\,\mathrm{Emb}_F(E, \overline{F}) = [E:F]$

E/F is splitting $\Leftrightarrow$ $\mathrm{Emb}_F(E, \overline{F}) = \mathrm{Aut}(E/F)$.

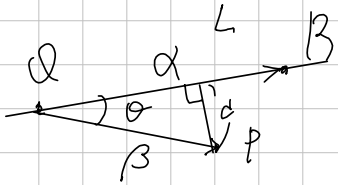
$\alpha, \beta \in \mathbb{R}^3$, $\gamma \in \mathbb{R}^3$,



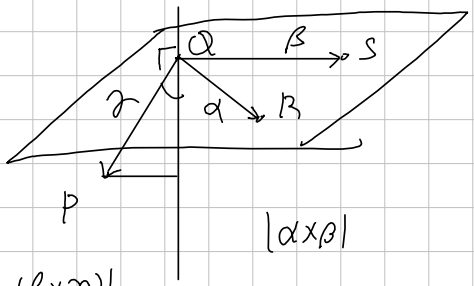
1. $|\alpha \times \beta| = |\alpha| \cdot |\beta| \cdot \sin\theta$.
2. $\alpha \cdot \beta = |\alpha| \cdot |\beta| \cdot \cos\theta$.
3. $V = A \cdot h = \underbrace{|\alpha| \cdot |\beta| \cdot \sin\theta}_{|\alpha \times \beta|} \cdot \underbrace{h}_{|\gamma| \cdot \cos\phi}$

$= |\alpha \times \beta| \cdot |\gamma| \cdot \cos\phi$

$= |\gamma \cdot (\alpha \times \beta)|$.



$$d = \frac{|\alpha \times \beta|}{|\alpha|}$$



$$d = \frac{|\alpha \cdot (\beta \times \gamma)|}{|\alpha \times \beta|}$$

$L: \mathbb{R} \to \mathbb{R}^3 : t \mapsto (a \;\; b \;\; c) \cdot (t) \left(= \begin{pmatrix} at \\ bt \\ ct \end{pmatrix} \right)$

$T: \mathbb{R}^3 \to \mathbb{R} : (x, y, z) \mapsto (a \;\; b \;\; c) \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$n = (a, b, c)$, and $P_0 = (x_0, y_0, z_0)$.

$n \cdot (P_0 - P) = 0 \iff a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$

1). $\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$ does not exist.

Proof. Take $f(x,y)=\frac{x^2-y^2}{x^2+y^2}$.

First, observe that $\lim_{y\to 0} f(0,y)=-1$

but, $\lim_{x\to 0} f(x,0)=1$. So that proved.

2). $f(x,y)=\frac{xy}{x^2+y^2}$

observe that $\lim_{x\to 0} f(x,x)=\frac{1}{2}$

but $\lim_{x\to 0} f(x,0)=0$ $\quad a+b\geq 2\sqrt{ab}$

3). $f(x,y)=\frac{x^2y}{x^2+y^2}$

Given $\varepsilon>0$, take $\delta=\varepsilon$. Then, $\|(x,y)\|<\delta \Rightarrow$

$$|f(x,y)-0|=\left|\frac{x^2y}{x^2+y^2}\right|\leq\frac{|x^2y|}{2|xy|}=\frac{1}{2}\cdot|x|<\delta=\varepsilon.$$

$\{X^1, X^2, \cdots, X^p\}$ is linearly indep-

$\Leftrightarrow$

$\sum_{k=1}^{p} a_k \cdot X^k = 0 \Rightarrow a_1 = a_2 = \cdots = a_p = 0$

$$\begin{pmatrix} X^1 & \cdots & X^p \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_p \end{pmatrix} = u_1 X^1 + \cdots + u_p X^p$$

assume $a_1 X^1 + \cdots + a_q X^q = 0$. since

$= a_1 X^1 + \cdots + a_q X^q + 0 \cdot X^{q+1} + \cdots + 0 \cdot X^p$ 이

$X\mu$

$X\mu = 0$ 인 μ가 $\mu = 0$ 으로 유일하다

$(\mu_1 + \mu_3)x + (\mu_1 + \mu_2)y + (\mu_2 + \mu_3)z = 0$

$$\begin{cases} \mu_1 + \mu_3 = 0 \\ \mu_1 + \mu_2 = 0 \\ \mu_2 + \mu_3 = 0 \end{cases}$$

$\mu_3 - \mu_2 = 0$

$2 \cdot \mu_3 = 0$

$\cdot \mu_3 = 0$

2=0

이

$1 + 1 = 0$

F_2

$f: \mathbb{R}^2 \to \mathbb{R}$ is differentiable at $(x,y)=(a,b)$

$\Leftrightarrow \lim_{h \to 0} \dfrac{|f((a,b)+h) - f(a,b) - \mathbb{B} \cdot h|}{\|h\|} = 0$

$f = \frac{1}{2}x^2 + 4xy + h(y)$,

$f_y = 4x + h'(y)$

$Z - Z_0 = f_x(x - x_0) + f_y(y - y_0)$

EXAMPLE 3. Define $f : \mathbf{R}^2 \to \mathbf{R}$ by setting $f(\mathbf{0}) = 0$ and

$$f(x, y) = x^2 y/(x^4 + y^2) \quad \text{if} \quad (x, y) \neq \mathbf{0}.$$

We show all directional derivatives of f exist at $\mathbf{0}$, but that f is not differentiable at $\mathbf{0}$. Let $\mathbf{u} \neq \mathbf{0}$. Then

$$\frac{f(\mathbf{0} + t\mathbf{u}) - f(\mathbf{0})}{t} = \frac{(th)^2(tk)}{(th)^4 + (tk)^2} \frac{1}{t} \quad \text{if} \quad \mathbf{u} = \begin{bmatrix} h \\ k \end{bmatrix}$$

$$= \frac{h^2 k}{t^2 h^4 + k^2},$$

so that

$$f'(\mathbf{0}; \mathbf{u}) = \begin{cases} h^2/k & \text{if } k \neq 0, \\ 0 & \text{if } k = 0. \end{cases}$$

$\Delta Z = f(a + \Delta x, b + \Delta y) - f(a, b)$

$dZ = f_x(x, y)\,dx + f_y(x, y)\,dy$

$A \in M_{n\times m}$ $\quad A = \begin{pmatrix} A_{11} & \cdots & A_{1m} \\ \vdots & & \vdots \\ A_{n1} & & A_{nm} \end{pmatrix}$ $\quad L_A : \mathbb{R}^m \to \mathbb{R}^n : x \mapsto Ax$

$Col(A) \subseteq \mathbb{R}^n$

$N(A) \subseteq \mathbb{R}^m = \{A$에 곱해지는 벡터$\}$ 집합

$m = \text{rank } A + \underset{\text{nullity } A}{\dim N(A)}$

A $\quad n$ by m

A^TA $\quad m$ by m

$m = \text{rank}A + \text{nullity } A$

$m = \text{rank}A^TA + \text{nullity}A^TA$

(포) $Ax = 0$

$A^T(Ax) = 0$

$A^T(0) = 0$

Ax

$x+1$

$\|$

1

$x \in N(A) \Rightarrow x \in N(A^TA)$

$A^TAx = 0$

$N(A) \subseteq N(A^TA)$

N ✳

$A^TAx = \vec{0}$

$0 = \vec{0} \cdot \vec{x}$

$= (A^TAx) \cdot x$

$= \|Ax\|^2$

$= 0 \quad Ax = 0$

$$\begin{pmatrix} A_1 \\ \vdots \\ A_n \end{pmatrix}$$

$$\begin{pmatrix} & \end{pmatrix}$$

A n by m

$N(A) = N(A^TA)$

$\subseteq$ ← General

$\supseteq$ ← Inner product. $A^TAx=0$

$\Rightarrow \|Ax\|=0 \Leftrightarrow Ax=0$

By dim. Thm, $m = \text{rank}\,A + \dim N(A)$

$= \text{rank}\,A^TA + \dim N(A^TA)$

Trank $A = \text{rank}\,A^t$

proof.

$(A^T)^T = A$

$a^2+b^2=0$

$\Rightarrow a=b=0$

$\text{rank}\,A^T \subseteq \text{rank}(A^T)^T = \text{rank}\,A$

$(A^T)^T = A$

$\sum x_i^2$

$\langle Ax, Ax \rangle$

$\langle A^TAx, x \rangle$

$x \cdot y = x^T \cdot y$

$A^TAx \cdot x = x^TA^TAx$

$\langle x, y \rangle = y^Tx.$

$\|x\|$

$= \sqrt{\sum_1^n (x_i)^2} = \sqrt{x \cdot x}$

$\langle x, y \rangle$

$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

$X \cdot X = X^TX = (x_1 \cdots x_n)\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{i=1}^n x_i^2$

19. 2. (1)

$$\lim_{(x,y,z)\to(0,0,0)} \frac{xe^{y}z^{2}-\cos(x^{2})yz}{x^{2}+3y^{2}+2z^{2}} \quad \text{exists?}$$

Proof. observe: for $(x,y,z)=(0,0,z)$ $(z\neq 0)$, it has zero.

But, let $(x,y,z)=(0,t,t)$ Then,

$$= \frac{-t^{2}}{5t^{2}} = -\frac{1}{5} \quad \text{for any } t\neq 0,$$

18. 2. (1) Determine:

$$\lim_{(x,y)\to(0,0)} \frac{x^{3}y^{2}\sqrt{x^{2}+y^{2}}}{x^{6}+y^{2}}$$

Since $x^{6}+y^{2} \geq 2\sqrt{x^{6}y^{2}} = 2|x|^{3}\cdot|y|$,

$$0 \leq \left|\frac{x^{3}y^{2}\sqrt{x^{2}+y^{2}}}{x^{6}+y^{2}}\right| \leq \frac{1}{2}\cdot y\cdot\sqrt{x^{2}+y^{2}} \to 0$$

or, $\frac{1}{x^{6}+y^{2}} \leq \frac{1}{y^{2}}$, so $\leq x^{3}\sqrt{x^{2}+y^{2}}$

Tip. force $\to 0$.

18. 3. $\lim \frac{xy+yz^{2}+xz^{2}}{x^{2}+y^{2}+z^{4}}$.

$(x,y,z)=(0,0,t)$

and $(t^{2},0,t)$

$$\lim \frac{y^{2}\cdot\cos x\cdot\sin y}{x^{2}+y^{2}}.$$

$$\left|\frac{y^{2}\cos x\sin y}{x^{2}+y^{2}}\right| \leq |\sin y| \to 0.$$

$f: \mathbb{R}^2 \to \mathbb{R}$ has critical value $\Leftrightarrow H(f) \geq 0$ or $H(f) \leq 0$.

$f(x,y) = (2x^2+y^2)\cdot e^{-(x^2+y^2)}$

$\nabla f = (f_x, f_y)$, $f_x = (4x - 4x^3 - xy^2)\cdot e^{-(x^2+y^2)}$

$f_y = ($

$f(x,y) = 4xy + 2x^2y - xy^2$

$f_x = 4y + 4xy - y^2$, $f_y = 4x + 2x^2 - 2xy$

$H(f) = \begin{pmatrix} 4y & 4+4x-2y \\ // & 4+4x-2y \end{pmatrix}$

[Preliminary.]

Let A be a m by n matrix.

Notation:

the Null space (or kernel) is $N(A) = \{X \in \mathbb{R}^n \mid AX = 0 \in \mathbb{R}^m\} (= \ker A)$

the Column space (or image) is $col(A) = \{AX \in \mathbb{R}^m \mid X \in \mathbb{R}^n\} (= \mathrm{Im}\, A)$.

And, $\begin{cases} \text{rank } A := \dim \mathrm{Im}\, A \\ \text{nulity } A := \dim \ker A. \end{cases}$

Clearly, $\text{rank } A \le m$ and $\text{nulity } A \le n$. And, by the dimension thm,

$n = \text{rank } A + \text{nulity } A$ implies $\begin{cases} \text{nulity } A = 0 \Rightarrow m \ge n. \\ \text{rank } A = m \Rightarrow m \le n. \end{cases}$

[1]. $A \in M_{m\times n}$ has a left inverse $\Leftrightarrow N(A) = \{0\}$.

Proof. Assume that $A^{-1} \in M_{n\times m}$ is a left inverse of A. Then,

$Ax = 0 \Rightarrow A^{-1}Ax = x = A^{-1}0 = 0$, so $x = 0$. Thus $\ker A = \{0\}$.

Conversely, assume A is non-singular. Then, for the standard basis $\mathcal{B} = \{e_1, \cdots, e_n\}$ for $\mathbb{R}^n$, $\{Ae_1, \cdots, Ae_n\}$ is linearly independent.

Now, since $m \ge n$, Consist a matrix

$$B = \begin{pmatrix} Ae_1 & \cdots & Ae_n & v_{n+1} & \cdots & v_m \end{pmatrix}$$

Where

$(A^TA)^{-1}A^T \cdot A = I$.

$A \cdot A^T(AA^T)^{-1} = I$.

$col(A) \simeq \mathbb{R}^m$

$col(A) \subseteq \mathbb{R}^m$

Ax

$Y \in \mathbb{R}^m$

$Y = AA^{-1}Y$

$= A(A^{-1}Y) \in col(A)$

$f^{-1}(f(x)) = x$. $x = f(u)$, $dx = f'(u)\,du$.

$\frac{1}{u}\int_1^3 u \cdot f(u)\,du$. $g'(f(u)) \cdot f'(x) = 1$.

$\int_0^2 2^{4u}$.

$\text{Row}(AB) = \langle \{e_1^T AB, \cdots, e_n^T AB\} \rangle$

$\text{Row}(B) = \langle \{e_1^T B, \cdots, e_n^T B\} \rangle$

since $\{e_1^T A, \cdots, e_n^T A\} \subseteq \{e_1^T, \cdots, e_n^T\}$, proven.

$$A = \begin{pmatrix} a_{11} & & 0 \\ & \ddots & \\ ? & & a_{nn} \end{pmatrix} \qquad \det A = a_{11} \cdots a_{nn}$$

$$A = \begin{pmatrix} a_{11} & & 0 \\ & \ddots & \\ 0 & & a_{nn} \end{pmatrix} + \begin{pmatrix} 0 & & 0 \\ & \ddots & \\ ? & & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & & \\ & \ddots & \\ a_{11} & & 1 \\ & \ddots & \\ & & a_{nn} \end{pmatrix}$$ nilpotent

$$A = I - N \qquad N^n = 0$$

$$(I + N + N^2 + \cdots + N^{n-1})$$

$$(I - N)(I + N + \cdots + N^{n-1})$$

$$= I + \cancel{N + \cdots + N^{n-1}} - \cancel{N - N^2 - \cdots - N^{n-1}} - \cancel{N^n}$$

$$= I$$

$$x^2 - 1 = (x-1)(x+1)$$

$$x^3 - 1 = (x-1)(x^2 + x + 1)$$

$$x^n - 1 = (x-1)\left(\sum_{i=0}^{n-1} x^i\right)$$

N^n

[If $x_n \to x$ and $y_n \to y$, then $x_n y_n \to xy$.

Proof. Since $x_n \to x$, for $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow |x_n - x| < 1$$

By the triangle inequality, for $n \geq N$,

$$|x_n| \leq |x| + |x_n - x| < |x| + 1$$

Now, for all $n \in \mathbb{N}$,

$$|x_n| < \max\{|x_1|, \cdots, |x_{N-1}|, |x| + 1\} + 1$$]

Therefore, $\{x_n\}$ is bounded by $M = \max\{|x_1|, \cdots, |x_{N-1}|, |x|+1\} + 1 > 0$.

Now, let $\varepsilon > 0$ be given. Take $\varepsilon' = \varepsilon / (M + |y|)$. Then, there are $N_1, N_2 \in \mathbb{N}$ such that

$$n \geq N_1 \Rightarrow |x_n - x| < \varepsilon' \quad \text{and} \quad n \geq N_2 \Rightarrow |y_n - y| < \varepsilon'.$$

And, for $n \geq \max\{N_1, N_2\}$,

$$\begin{aligned} |x_n y_n - xy| &= |x_n y_n - x_n y + x_n y - xy| \\ &\leq |x_n| \cdot |y_n - y| + |y| \cdot |x_n - x| \\ &< M \cdot \varepsilon' + |y| \cdot \varepsilon' \\ &\leq (M + |y|) \cdot \frac{\varepsilon}{M + |y|} = \varepsilon \end{aligned}$$

So, proved. □